FIITJEE Students From All Programs have bagged 33 in Top 100, 77 in Top 200 and 205 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2015:
4551 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2015.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE(Advanced)-2016

ANSWERS, HINTS & SOLUTIONS

PART TEST– IV
(Paper - 2)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	A, C, D	A, B	A, D
2.	C, D	A, C	B
3.	A, B, C	A, B, D	A, B, C, D
4.	C	B, C	B,C, D
5.	A	D	A
6.	C	A, B	A, B
7.	B	A, B	A, B, D
8.	B	B, C, D	A, B
9.	B	C	D
10.	A	A	A
11.	A	C	B
12.	B	A	B
13.	A	B	B
14.	D	A	C
15.	A	A	D
16.	B	A	A
17.	D	A	B
18.	A	D	C
19.	C	B	D
20.	B	C	B

HINTS AND SOLUTION

Physics

PART – I

SECTION – A

9-10. $evB = \frac{mv^2}{r}$

$mvr = \frac{nh}{2\pi}$

$\frac{mr}{eB} = \frac{nhr}{2\pi mv^2}$

$m^2v^2 = \frac{eBnh}{2\pi}$

$K.E = \frac{1}{2}m\frac{eBnh}{2\pi m}$

$K.E. = \frac{eBnh}{4\pi m}$

$r = \frac{nh}{2\pi m}\sqrt{\frac{2\pi m^2}{eBnh}} = \sqrt{\frac{nh}{2\pi eB}}$

Chemistry

PART – II

SECTION – A

1. Cataphoresis is the process used for coagulation of colloids in which +ve particles move towards –ve pole and –ve particles move towards +ve pole in the presence of electric field.

2. SiO_2 has different molecular arrangements in different forms.

4. More the value of *i* lesser will be freezing point of solution and $i = \frac{\text{Molecular wt. theoretical}}{\text{Molecular wt. experimental}}$

5. The first statement is false because F_2 and Cl_2 are gases, Br_2 is liquid while I_2 is a solid. Bromine dissolves in chloroform imparting reddish-brown colour. Chlorine dioxide is evolved when $KClO_3$ is treated with conc. H_2SO_4

6. Surface tension of lyophilic aquasols is lower than that due to force of attraction between dispersed phase and dispersion medium.
Its viscosity is higher due to force of attraction between dispersed phase and dispersion medium.

13. $\Delta G = -nFE$
$= -2 \times 96500 \times 1.1$
$= -212300$ J

14. $(\Delta G°)_R = (G°)_P - (G°)_R$
$= -154 - 64.4 = -218.4$ kJ.
$\Delta G° = -nFE°$
$-218.4 \times 10^3 = -2 \times 96500 \times E°$
$E° = 1.13$ V.

15. Let the solution is 100 gm
$\therefore$ 25 gm is impure $MgCl_2$ ($MgCl_2 + MCl_3$)
$\therefore$ 6 gm is MCl_3

$$\Delta T_f = i \times k_f \times m$$

$$= 1.86 \times \left[\left(i \times \frac{w_B}{M_B}\right)_{MgCl_2} + \left(i \times \frac{w_B}{M_B}\right)_{MCl_3}\right] \times \frac{1000}{w_A}$$

$$= 1.86 \times \left[\left(3 \times \frac{19}{95}\right) + \left(2.5 \times \frac{6}{150}\right)\right] \times \frac{1000}{75}$$

$$= 1.86 \times (0.6 + 0.1) \times \frac{1000}{75}$$

ΔT_f = **17.36 K**

16. At –21.7°C amount of solvent can be calculated as

$$21.7 = 1.86 \times \left[\left(3 \times \frac{19}{95}\right) + \left(2.5 \times \frac{6}{150}\right)\right] \times \frac{1000}{w_A}$$

$w_A = 60$ gm
$\therefore$ Amount of ice formed = 75 – 60 = **15 gm**.

Mathematics PART – III

SECTION – A

1. The given lines are perpendicular for all values of λ and μ if the vectors $2i + pj + 5k$ and $3i - pj + pk$ are perpendicular
$\Rightarrow 2 \times 3 + p(-p) + 5p = 0$
$\Rightarrow p^2 - 5p - 6 = 0 = 0 \Rightarrow p = -1$ or 6

2. Equation of any plane through the line is $a(x - 1) + b(y + 1) + c(z - 3) = 0$
where $2a - b + 4c = 0$ ... (i)
The plane $x + 2y + z = 6$ perpendicular to $a(x - 1) + b(y + 1) + c(z - 3) = 0$
$\therefore\ a + 2b + c = 0$ ... (ii)
Solving (i) & (ii) we get $\frac{a}{-9}=\frac{b}{2}=\frac{c}{5}$
equation of the plane is $9x - 2y - 5z + 4 = 0$
Hence the equation of projection is $9x - 2y - 5z + 4 = 0$

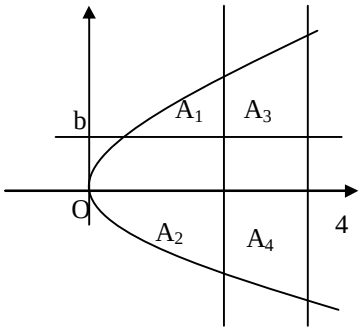
3. $P(A_1 \cap A_2) = 1/4$
$P(A_1) = 2/4$
$P(A_2) = \frac{1}{2}$. Thus $P(A_1 \cap A_2) = P(A_1)\,P(A_2)$. Similarly other are true.

4. $P(A) = 1/4$, $P(B/(A) = 5/7$, $P(B/\bar{A}) = 6/7$
$P(B) = P(A)\,P(B/(A) + P(\bar{A})\,P(B/\bar{A}) = 1/4 \times 5/7 + 3/4 \times 6/7 = 23/28$
$P(A/(B) = \frac{P(A \cap B)}{P(B)}=\frac{1/4\times5/7}{1/4\times5/7+3/4\times6/7}=\frac{5}{23}$
$P(A/\bar{B}) = \frac{P(A \cap \bar{B})}{P(\bar{B})}=\frac{\frac{1}{4}\times\frac{2}{7}}{1-\frac{23}{28}}=\frac{2}{5}$

5. $A^3 = B^3$ …….(i)
$A^2B = B^2A$.....(ii)
subtracting (ii) by (i)
$\left(A^2 + B^2\right)(A - B) = 0$
$\because |A - B| \neq 0$, $A^2 + B^2 = 0$

6. Apply $C_1 \to C_1 - C_2$ and $C_2 \to C_2 - C_3$
$$\Delta = \begin{vmatrix} -d & -d & a+2d \\ -d & 2d & a \\ 2d & -d & a+d \end{vmatrix} = -9(a+d)d^2$$

7. Total area = $2\int_0^4 \sqrt{x}\,dx = \frac{32}{3}$



Area of each part = 8/3

$A_3 = A_4 \Rightarrow \int_a^4 \left(\sqrt{x} - b\right)dx =$

$\int_a^4 \left(b + \sqrt{x}\right)dx = \frac{8}{3} \Rightarrow b = 0$

$\int_a^4 \sqrt{x}\,dx = \frac{8}{3} \Rightarrow a^3 = 16$

8. If y = f(x) is the curve, then

Y – y = f ′(x) (X – x) is the equation of the tangent of (x, y). Take f′ (x) = $\frac{dy}{dx}$

Putting X = 0, the initial ordinate of the tangent is = y – x f′(x) = subnormal at this point = $y\frac{dy}{dx}$ $\Rightarrow$ $y - x\frac{dy}{dx} = y\frac{dy}{dx}$

Which is homogenous of first degree

Again

$\Rightarrow$ = 1 which is linear in x

$\Rightarrow$ (A), (B) are correct.

9-10. $\vec{b} = -\frac{1}{2}\vec{a} + \frac{\sqrt{3}}{2}\hat{k} = -\frac{1}{2\sqrt{2}}\hat{i} - \frac{1}{2\sqrt{2}}\hat{j} + \frac{\sqrt{3}}{2}\hat{k}$

$8\left(\left(\frac{1}{2\sqrt{2}}\right)^4 + \left(\frac{1}{2\sqrt{2}}\right)^4 + \left(\frac{\sqrt{3}}{2}\right)^4\right) = \frac{19}{4}$

also if $|\vec{b} \times \vec{c}| = |\vec{a} \times \vec{c}|$, then $\vec{c}$ is perpendicular to both $\vec{a}$ and $\vec{b}$

$\Rightarrow \vec{c} = \hat{k} \times \left(\frac{\hat{i}}{\sqrt{2}} + \frac{\hat{j}}{\sqrt{2}}\right)$

$(\vec{b} \cdot \vec{c})_{max}$ = cos30° and $(\vec{b} \cdot \vec{c})_{min}$ = cos150°.

11. T is an equilateral triangle with the vertices at (1, 0, 0), (0, 1, 0) and (0, 0, 1)

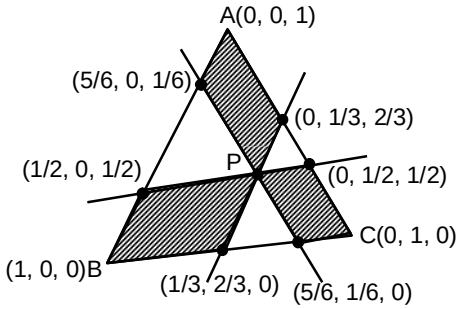
$\Rightarrow$ area of the region T is $\frac{\sqrt{3}}{2}$.

12. Take a point $P\left(\frac{1}{2}, \frac{1}{3}, \frac{1}{6}\right)$ on the plane x + y + z = 1

The region S is shown as shaded region.

Area of S = $\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{4}(a^2 + b^2 + c^2)$, where a, b, c are sides of three small equilateral triangles

= $\frac{7\sqrt{3}}{36}$.



13–14. Observe that H = PP^* (P^* = transposed conjugate of P)

So $\det(H) = \det(P) \cdot \overline{\det(P)}$

Also $|\det(H)| = |\det(P)|^2 \Rightarrow \det(H)$ is non negative integer

Now in matrix P perform $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$ and take common 1 – i (if possible) from R_2 and R_3

det(P) becomes $(1 - i)^2 z \Rightarrow \det(H) = 2^2|Z|^2$

15–16. f(x) and $f^{-1}(x)$ intersect at $f(x) = f^{-1}(x) = x$

so at $x = 2^{1/3}$

so area bounded is $(2^{1/3})^2 - 2\int_1^{2^{1/3}} f(x)\,dx = 3 \cdot 2^{1/3} - \frac{5}{2}$ so $t = 2^{1/3}$

so a = 2, b = 3, n = 15.

Last two digits of 2^{2012} are 96 $\Rightarrow$ difference = 3.

Number of ways of distribution of apples is coefficient of x^{15} in $(1 + x + x^2 + \ldots + x^5)^6$

Number of ways of distribution of oranges is 1.

17 (P) $\vec{a}, \vec{b}$ unit vectors

$$\left(\vec{a} + 2\vec{b}\right) \cdot \left(5\vec{a} - 4\vec{b}\right) = 0$$

$$\Rightarrow 5|\vec{a}|^2 + 6 \cdot \vec{a} \cdot \vec{b} - 8\left|\vec{b}\right|^2 = 0$$

$$\Rightarrow 6\vec{a} \cdot \vec{b} = 3$$

$$\Rightarrow \vec{a} \cdot \vec{b} = \frac{1}{2}$$

$$\Rightarrow 2\left(\vec{a} \cdot \vec{b}\right) = 1$$

(Q) $\vec{a} = (1,0,3), \vec{b} = (-1,3,4), \vec{c} = (1,2,1)$ and $\vec{d} = (k,2,5)$ are coplanar if

$$\left[\vec{d}\ \ \vec{b}\ \ \vec{c}\right] + \left[\vec{d}\ \ \vec{c}\ \ \vec{a}\right] + \left[\vec{d}\ \ \vec{a}\ \ \vec{b}\right] = \left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]$$

$$\Rightarrow \begin{vmatrix} k & 2 & 5 \\ -1 & 3 & 4 \\ 1 & 2 & 1 \end{vmatrix} + \begin{vmatrix} k & 2 & 5 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{vmatrix} + \begin{vmatrix} k & 2 & 5 \\ 1 & 0 & 3 \\ -1 & 3 & 4 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 3 \\ -1 & 3 & 4 \\ 1 & 2 & 1 \end{vmatrix}$$

$\Rightarrow k = -1.$

(R) (1, 1, m), (1, 1, m + 1), (1, –1, m) are coplanar if

$$\begin{vmatrix} 1 & 1 & m \\ 1 & 1 & m+1 \\ 1 & -1 & m \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} 1 & 0 & m \\ 1 & 0 & m+1 \\ 1 & -2 & m \end{vmatrix} = 0$$

$\Rightarrow 2 = 0$ which is absurd.

$\therefore$ there is no value of m for which the vectors are coplanar.

(S) $\vec{a}\times(\vec{b}\times\vec{c})=(\vec{a}\cdot\vec{c})\vec{b}-(\vec{a}\cdot\vec{b})\vec{c}$

$\vec{b}\times(\vec{c}\times\vec{a})=(\vec{b}\cdot\vec{a})\vec{c}-(\vec{c}\cdot\vec{b})\vec{a}$

$\vec{c}\times(\vec{a}\times\vec{b})=(\vec{c}\cdot\vec{b})\vec{a}-(\vec{c}\cdot\vec{a})\vec{b}$

since dot product is commutative, so we have, the sum = 0

18. (P) Required probability $=\frac{1}{2}$.

(Q) Required probability $=\left(\frac{1}{2}\right)^3+\left(\frac{1}{2}\right)^4+\left(\frac{1}{2}\right)^4+\left(\frac{1}{2}\right)^4+\left(\frac{1}{2}\right)^4-\left(\frac{1}{2}\right)^7=\frac{47}{2^7}$.

(R) Required probability $=\frac{4}{2^7}$.

(S) There are five cases 0, 1, 2, 3 or 4 wins

Required probability $=\frac{1+7+15+10+1}{2^7}=\frac{34}{2^7}$.

19. $\Delta=\begin{vmatrix}a^2 & a^2-(b-c)^2 & bc\\ b^2 & b^2-(c-a)^2 & ca\\ c^2 & c^2-(a-b)^2 & ab\end{vmatrix}$ $\quad C_2\to C_2-C_1$

$=-\begin{vmatrix}a^2 & (b-c)^2 & bc\\ b^2 & (c-a)^2 & ca\\ c^2 & (a-b)^2 & ab\end{vmatrix}$ $\quad C_2\to C_2+2C_3$

$=-\begin{vmatrix}a^2 & b^2+c^2 & bc\\ b^2 & c^2+a^2 & ca\\ c^2 & a^2+b^2 & ab\end{vmatrix}=-(a^2+b^2+c^2)\begin{vmatrix}1 & b^2+c^2 & bc\\ 1 & c^2+a^2 & ca\\ 1 & a^2+b^2 & ab\end{vmatrix}$ $\quad C_1\to C_1+C_2$.

$=-(a^2+b^2+c^2)\begin{vmatrix}0 & (b-a)(b+a) & c(b-a)\\ 0 & (c-b)(c+b) & a(c-b)\\ 1 & a^2+b^2 & ab\end{vmatrix}$ $\quad R_1\to R_1-R_2,\ R_2\to R_2-R_3$

$=-(a^2+b^2+c^2)(b-a)(c-b)(a+b+c)$

$=(a-b)(b-c)(c-a)(a+b+c)(a^2+b^2+c^2)$

$\Delta=(a-b)(b-c)(c-a)(a+b+c)(a^2+b^2+c^2)$

(P) If $a+b+c=0 \Rightarrow \Delta=0$

Given system is homogeneous system, therefore infinite number of solutions will exist.

(Q) If $a=b=c=0$, then all three equation will be $0\cdot x+0\cdot y+0\cdot z=0$

therefore any (x, y, z) will satisfy the given set of equations where $x, y, z\in R$.

(R) $a=b=c,\ a+b+c\neq 0$

then all three equations will come in form $x+y+z=0$

Solution of equation are $(\lambda_1, \lambda_2, -\lambda_1-\lambda_2)$

(S) $\Delta\neq 0$

$\Rightarrow$ only (0, 0, 0) is the solution. $\Rightarrow$ unique solution.